

Case 06: Two-Dimensional Firebrand-Driven Spread

1 Verification purpose

Case ID: CASE06_F2D

This case verifies that Eulerian spotting produces a genuinely two-dimensional, firebrand-driven fireline under a manufactured two-component, space–time wind. It exercises emission, vector transport, deposition, direct spot ignition, and level-set propagation together.

The test exposes loss of either wind component, incorrect direction conversion, one-dimensional source geometry, missing secondary ignitions, non-monotone burned area, and stale transient output. Success requires nested perimeters near 30, 60, 90, and 120 s that advance in both directions and exceed the surface-only reach.

2 Mathematical formulation and numerical implementation

The prescribed horizontal wind is

$$u(x, t) = \bar{u} + A \sin(2\pi x/\lambda_x) \sin(2\pi t/T), \quad (1)$$

$$v(y, t) = \bar{v} + B \sin(2\pi y/\lambda_y) \sin(2\pi t/T), \quad (2)$$

with $\bar{u} = 6.71 \text{ m s}^{-1}$, $A = 3.355 \text{ m s}^{-1}$, $\bar{v} = 1.0 \text{ m s}^{-1}$, $B = 0.5 \text{ m s}^{-1}$, $\lambda_x = \lambda_y = 100 \text{ m}$, and $T = 60 \text{ s}$. The preprocessor evaluates these equations at every cell centre and every one-second meteorology time. It writes $|\mathbf{u}|$ in mph and converts the transport-to angle to ELMFIRE meteorological wind-from convention.

For each Eulerian source tracker, the current spotting implementation advances

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \mathbf{u}(\mathbf{x}_k, t_k) \Delta t_k, \quad \Delta t_k = \Delta x / \max(|\mathbf{u}|, 0.01),$$

and distributes the source ember count through the configured downwind lognormal landing distribution. Direct ignition updates the level-set state, after which both the primary headfire and spot fires create new burning cells. The configured surface reference speed is $R_s = 0.71 \text{ m s}^{-1}$ [1], and the corresponding emission residence time is $t_e = \Delta x / R_s = 14.08 \text{ s}$.

The prescribed u and v components determine an individual transport characteristic, but they do not uniquely determine the maximum coordinates of the coupled fireline in this configuration. The observed $\phi = 0$ perimeter is the union of the surface-propagated primary fire and surface-propagated spot fires created after nonlocal, probability-distributed deposition. Consequently, $x_{\max}$ and $y_{\max}$ do not generally satisfy the uncoupled equations $dx_{\max}/dt = u(x_{\max}, t)$ and $dy_{\max}/dt = v(y_{\max}, t)$. An exact endpoint metric would require an independent solution of the complete coupled problem.

3 Derivation of expected behavior

Because the wind has positive mean components, successive fronts must progress downwind and develop a nonzero crosswind span. The level-set burned set is irreversible, so for burned masks $B_j = \{(x, y) : \phi(x, y, t_j) \leq 0\}$ the expected inclusion is $B_j \subseteq B_{j+1}$. A conservative surface-only leading-edge bound is

$$x_{\text{surf}}(120) = x_0 + R_s(120) = 90.2 \text{ m}.$$

A front substantially beyond this location, together with nonzero ember-ignition cells, demonstrates that firebrands rather than the surface headfire alone control the long-range advance. Accordingly, the principal expected result is a family of nested two-dimensional contours rather than a one-dimensional TOA curve.

The largest generated transport speed is

$$|\mathbf{u}|_{\max} = \sqrt{(6.71 + 3.355)^2 + (1.0 + 0.5)^2} = 10.1762 \text{ m s}^{-1},$$

so preprocessing sets $\Delta t = 0.5(10)/10.1762 = 0.491344 \text{ s}$. The surface residence time is independently

$$t_e = 10/0.71 = 14.0845 \text{ s}.$$

Starting from $x_0 = 5 \text{ m}$, a surface-only head fire would be at

$t \text{ (s)}$	30	60	90	120
$x_{\text{surf}} \text{ (m)}$	26.3	47.6	68.9	90.2

and would advance $0.71(30) = 21.3 \text{ m}$ between successive target times. The declared 20-m interval threshold is therefore slightly below the surface-only analytical increment but still requires two resolved cells. At 120 s, the firebrand-excess test $x_{\max} - 90.2 \geq 30 \text{ m}$ is equivalently $x_{\max} \geq 120.2 \text{ m}$, together with a positive ember-ignition count. Exact irreversible evolution gives $|B_j \setminus B_{j+1}|/|B_j| = 0$; the numerical allowance is 0.02. The final span reference is qualitative nonzero two-dimensional growth, made quantitative as at least $40 \text{ m} = 4\Delta x$.

4 Verification metrics and acceptance criteria

The postprocessor selects transient phi rasters within 2 s of all four target times and excludes the two-cell halo. It calculates: (i) completeness of all four contours; (ii) the fraction $|B_j \setminus B_{j+1}|/|B_j|$, limited to 0.02; (iii) downwind leading-edge advance, required to be at least 20 m in every interval; (iv) final crosswind span, required to be at least 40 m; and (v) firebrand excess advance $x_{\max}(120) - x_{\text{surf}}(120)$, required to be at least 30 m with at least one ember-ignition cell. The overall result is PASS only when every component passes.

4.1 Justification of metric quantification

The thresholds are expressed in grid-resolvable units. Selecting a contour within 2 s limits time mismatch to less than 7% of the earliest 30 s target and only a few solver steps. Exact level-set evolution is nested, so the 2% burned-area allowance is reserved for zero-contour classification along a cell-wide perimeter; a larger loss would indicate physical unburning or inconsistent dump selection. The minimum interval advance of 20 m, final crosswind span of 40 m, and firebrand excess of 30 m equal two, four, and three cells, respectively. These multi-cell thresholds cannot be satisfied by a stationary front, a single misclassified boundary cell, or the surface-only reference, yet they do not prescribe a detailed perimeter shape that the manufactured problem does not supply. At least one ember-ignition cell is an exact mechanism check, and all four times are required to support the claimed two-dimensional evolution.

4.2 Predeclared acceptance contract

Table 1 summarizes the metrics fixed before inspection of the calculated results. Numerical values and component statuses are reported separately in Section 7.

Table 1: Predeclared verification metrics and acceptance criteria.

Metric	Output, region, and aggregation	Acceptance criterion
Perimeter completeness	Unique transient phi raster selected within 2 s of 30, 60, 90, and 120 s.	4 of 4 target times
Nesting violation	Maximum $ B_j \setminus B_{j+1} / B_j $ over consecutive physical burned masks.	≤ 0.02
Downwind progression	Increase in maximum burned-cell x between consecutive target times.	≥ 20 m every interval
Final crosswind span	$\max(y) - \min(y)$ of the final physical burned mask.	≥ 40 m
Firebrand-driven excess	$x_{max}(120) - [x_0 + R_s(120)]$, together with the final ember-ignition map.	≥ 30 m and at least one ember-ignited cell
Overall decision	Conjunction of completeness, nesting, progression, span, and excess-advance checks.	PASS only if all pass

5 Simulation configuration and predicted outcome

Figure 1 records the prepared spatial inputs for the base two-dimensional run. The postprocessor reads these panels from the variant-local GeoTIFF files, removes the two-cell numerical buffer, and retains map coordinates in metres. The left panel shows the most informative fuel or structure layout available for the case; the right panel shows the cells selected by the non-positive initial level-set field, making a localized ignition or firebrand-emission source explicit.

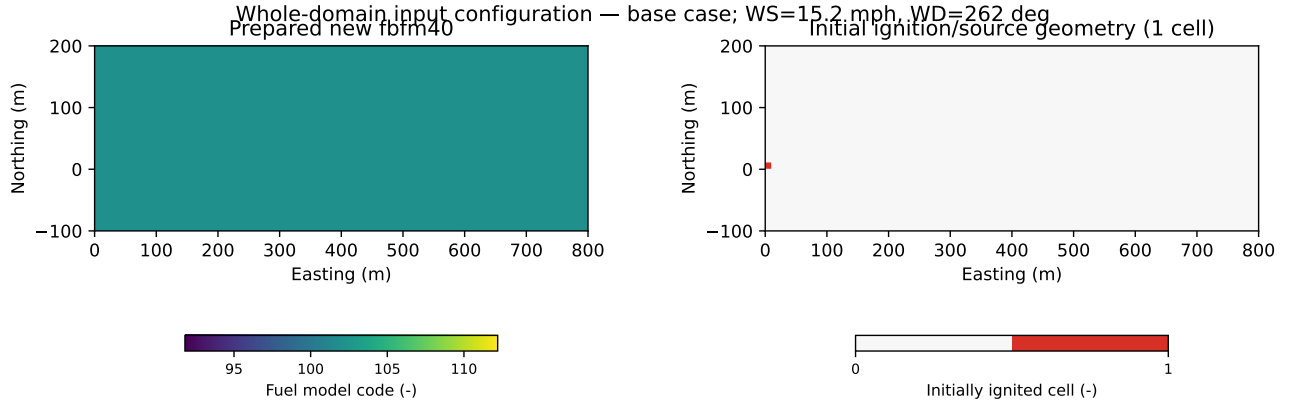


Figure 1: Whole-domain prepared input configuration for the representative variant. The physical domain is shown without the numerical halo; coordinates, configured wind values, fuel or structure layout, and initial ignition/source geometry are identified in the panels.

Before inspecting results, the predicted output is four nested contours elongated in positive x , displaced and broadened in y , with many secondary ember ignitions and a final leading edge far beyond 90.2 m.

6 Actual simulation results

Figure 2 shows the representative accumulated firebrand density from the base two-dimensional run. It is read from the actual ELMFIRE output named in the panel, masks nodata, and excludes the two-cell

Table 2: Configured two-dimensional transport case experiment.

Parameter	Value	Role
Physical domain	800×300 m	two-dimensional propagation
Cell size	10 m	reference spatial resolution
Raster grid	84×34 cells	includes two-cell halo
Meteorology	121 bands at 1 s	space-time wind
Simulation step	0.491 026 s	wind CFL approximately 0.5
Initial ignition	(5, 5) m	localized upwind source
Surface headfire	enabled	primary moving source
Generation	PER-AREA, 10 pcs $\text{m}^{-2} \text{s}^{-1}$	deterministic ember supply
Accumulation / ignition	Eulerian / direct	tested implementation path
Crosswind PDF	$\mu_y = 0, \sigma_y = 0.1$ m	narrow $k_{\max} = 0$ limit
Output times	30, 60, 90, 120 s	transient perimeter comparison

numerical buffer. Counts are divided by physical cell area, giving pcs m^{-2} and exposing the full lateral as well as downwind distribution.

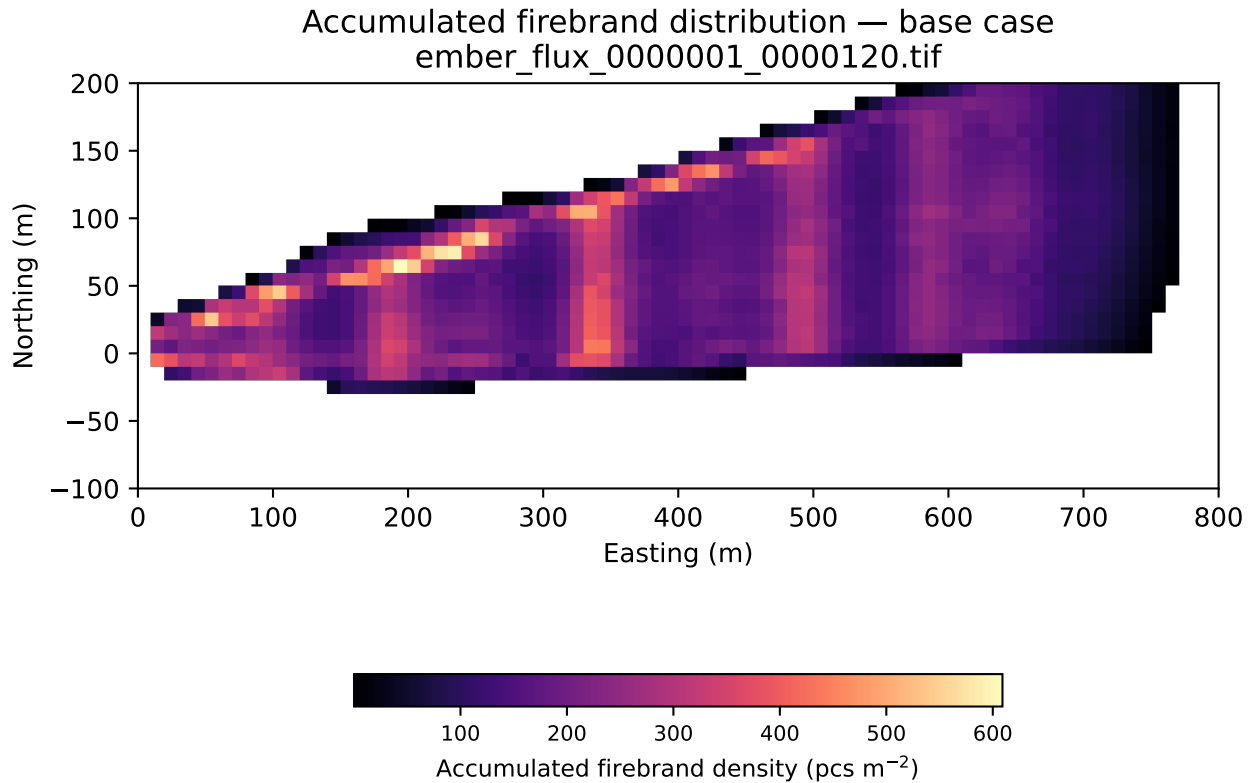


Figure 2: Whole-domain accumulated firebrand density from the representative completed ELMFIRE run. This domain-scale result supplements, rather than replaces, the higher-level verification profiles, convergence plots, ensemble statistics, and calculated acceptance metrics below.

The selected transient perimeter times were 30.44, 60.4, 90.35, 120 s. Exact filenames are recorded in outputs/metrics.json. Figure 3 first presents the evolving zero-level contours, which are the principal two-dimensional reference observable.

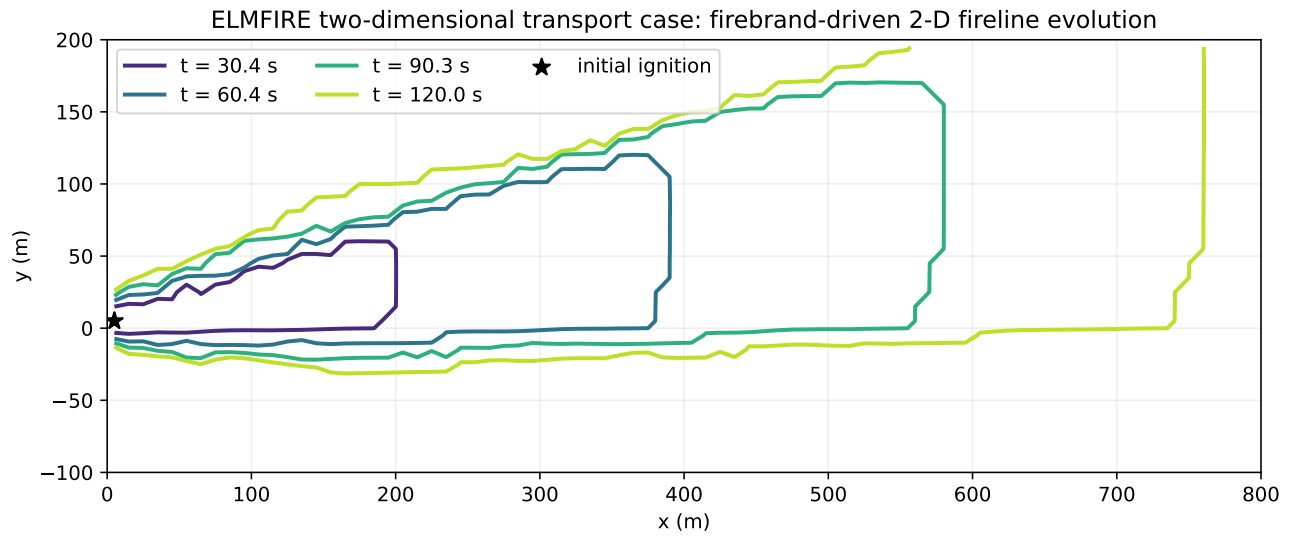


Figure 3: Actual ELMFIRE fire perimeters at four output times under the prescribed two-component, space-time wind. The star marks the initial ignition.

Figure 4 identifies the cells actually ignited by deposited firebrands. The accumulated-firebrand field is not repeated here because its whole-domain distribution is already presented in Figure 2.

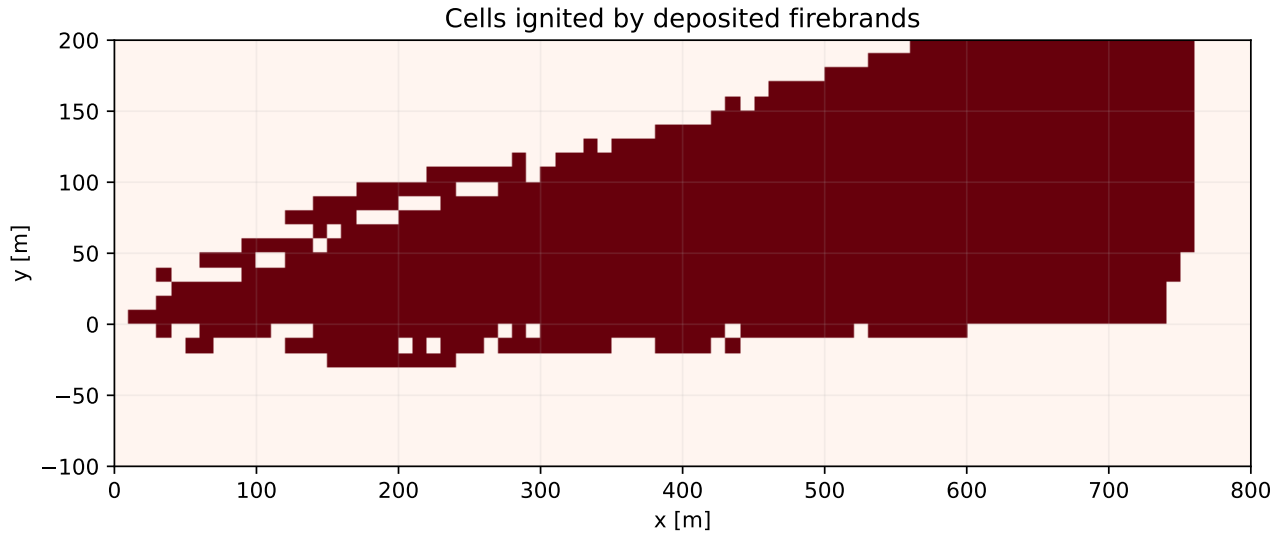


Figure 4: Whole-domain cells marked by ELMFIRE as ignitions caused by deposited firebrands. The numerical halo is excluded.

Table 3: Acceptance metrics calculated from actual ELMFIRE outputs.

Metric	Acceptance limit	Calculated value	Status
Required contours	4 within 2 s	30.44, 60.4, 90.35, 120	PASS
Nesting violation	≤ 0.02	0.0	PASS
Interval advances	each ≥ 20 m	190, 190, 180 m	PASS
Final crosswind span	≥ 40 m	220.0 m	PASS
Firebrand excess	≥ 30 m and ignitions	664.8 m; 1097 cells	PASS
Overall	all components true	PASS	pass

7 Calculated metrics and verification decision

8 Technical interpretation

The result is **pass**. The maximum nesting violation is 0.0, while the leading edge advances by 190, 190, 180 m over the three intervals. The final crosswind span is 220.0 m, confirming that the calculation is not a replicated one-dimensional transect. The final front exceeds the surface-only bound by 664.8 m and the ember-ignition raster contains 1097 positive cells; together these observations attribute the long-range evolving fireline to secondary ignitions from transported firebrands.

9 Scope and limitations

This test verifies quantified mechanism and evolution invariants of the current Eulerian 2-D coupling and reconstructs the intended contour diagnostic. The known wind components alone are not treated as an exact outer-fireline solution because surface spread and nonlocal secondary ignitions remain active. The test does not reproduce the comparison solution cell-for-cell, compare Eulerian and Lagrangian contours, validate the landing PDF against experiments, or prove grid convergence. The configured flat terrain isolates horizontal transport; additional terrain-slope cases would be required to verify topographic coupling.

References

- [1] Y. Qin, *A Physics-Based Eulerian Framework for Modeling Firebrand Showering in Regional-Scale Wildland and Wildland–Urban Interface Fire Simulations*, Ph.D. dissertation, University of Maryland, College Park, 2025.